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Uttarakhand incidence: Disaster, natural but facilitated by human beings?

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Abstract : Disaster Management is the process of addressing an event that has the potential to seriously disrupt the social fabric of the community. Disaster management is similar to disaster mitigation; however, it implies a whole of government approach to using community resources to fight the effects of an event and assumes the community will be self-sufficient for periods of time until the situation can be stabilized. Through disaster management, we cannot completely counteract the damage, but it is possible to minimize the risk through early warning, provide development plans for recuperation from the disaster, generate. Communication and medical resources, aid in rehabilitation and post-disaster reconstruction.

The exchange of correct information following the event is important, in order to ensure the resources necessary to support response and recovery activities. The 72 hours following a major event is the most difficult time due to lack of coordination among relief organizations.

Key words: Disaster mitigation, Natural disaster, Sustainable development, Disaster response, Rampant Mining.

INTRODUCTION

Disaster can happen at any time and any place; their human and financial consequences are hard to predict. The number of disaster each year is increasing, but only 50% of events get assistance. In,Yokohama (May 1994) the declaration of International decade for natural disaster reduction, underlined the need for an emphatic shift in the strategy for disaster mitigation (Yokohama, 1994). It was inter-alia stressed that disaster prevention, mitigation; preparedness and relief are four elements which contribute to and gain, from the implementation of the sustainable development policies. These elements along with environmental protection and sustainable development, are closely inter related. Therefore, nations should incorporate them in their development plans and ensure efficient follow up measures at the community, sub-regional, regional, national and international levels. It is well known that disaster prevention, mitigation and preparedness are better than disaster response in achieving the goals and objectives of vulnerability reduction. Disaster response alone is not sufficient as it yields only temporary results at a very high cost. Prevention and mitigation contribute to lasting improvement in safety and are essential to integrated disaster management.

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Mitigation

Mitigation is the effort to reduce loss of life and property by lessening the impact of disasters. Mitigation is taking action now or before the next disaster to reduce human and financial consequences later (analyzing risk, reducing risk, insuring against risk). Effective mitigation requires awareness and understanding of all local risks, address the hard choices and investment in long term community well-being. Without mitigation actions, we jeopardize our safety, financial security, and self reliance.

Disaster Mitigation

Disaster Mitigation measures are those, eliminate or reduce the impacts and risks of hazards through proactive measures taken before an emergency or disaster occurs.

Disaster preparedness starts at the local community level, if would branch out to the national level, and if needed, the international level.

Disaster mitigation is the ongoing effort to lessen the impact disasters have on people and property. Fewer people and communities would be affected by natural disasters with the use of this process. Because of the varying degree of each natural disaster, there are different mitigation strategies for each.

Some of the natural / manmade disasters are as follows:-

a) Earthquake
b) Extreme Heat
c) Flood
d) Hurricane
e) Landslides & Mudflows
f) Tornado
g) Tsunami
h) Volcano
i) Wild fire, etc.

Indian Government policy on Disaster Management & Mitigation

The Government of India has adopted mitigation and prevention as essential components of their development strategy. The 10th five year plan document has a detailed chapter on Disaster Management. The plan emphasizes the fact that development cannot be sustainable without mitigation being built into developmental process. Each state is supposed to prepare a plan scheme for disaster mitigation in accordance with the approach outlined in the plan. In short, mitigation is being institutionalized into developmental planning.

The Government of India has issued guidelines that where there is a shelf of projects, projects addressing mitigation will be given a priority. It has also been mandated that each project in a hazard prone area will have disaster prevention / mitigation as a term of reference and the project document has to reflect as to how the project addresses that term of reference.

The Disaster Management Act of 2005, under which the National Disaster Management Authority (NDMA) and the State Disaster Management Authority (SDMA) were setup, gives broad guidelines for disaster management. Every state has to frame its own Disaster Management Act keeping local conditions and dangers in mind.

NDMA was constituted in 2006 to lay down policies and guidelines for effective management, risk mitigation and prevention of disasters in the country. It is headed by the Prime Minister. According to the Disaster Management Act of 2004, NDMA should have an advisory committee of experts for disaster management at the national, state and district levels. In 2007, it constituted its first advisory committee for two years. The committee's term was extended for a year. After its end, setting up of a new committee was delayed because many ministries failed to nominate experts.

The NDMA is accused of taking up projects and leaving them incomplete. In 2008, NDMA initiated a nationwide project on floods and landslide mitigation. These projects have either been junked or have gone back to the drawing board midway. Similarly, a project to prepare vulnerability atlas of landslides, floods and earthquakes started five years ago. It is still incomplete.

In April 2013, CAG placed a report in Parliament pointing out that NDMA is not properly informed about the Disaster Management work in states. Projects that were initiated for disaster preparedness and mitigation have not been properly implemented. Worse, since 2010, the authority has been functioning without a core advisory committee of experts that advises it on different aspects of Disaster Management.

SDMA in Uttarakhand was formed in 2007. But five years later, it is yet to frame a disaster management plan, as reported by Comptroller and Auditor General (CAG) of India after auditing the SDMA. CAG also pointed out that only 66 of the 117 sanctioned posts in the state authority have been filled up.

Report of Mock Drills conducted by NDMA in May and June 2011 in Dehradun, Haridwar and Tehri Garhwal to test disaster preparedness in Uttarakhand, was not made public. It indicates that there was gaping holes in implementation of Disaster Management.

Rampant Mining on Riverbanks and Indiscriminate Construction of Hydropower Projects in Uttarakhand:

Ever since Uttarakhand was created in 2000, the State Government has been working with one agenda – exploit natural resources of water, forests and minerals to develop infrastructure, without caring for its consequences on nature, even knowing that the Himalayas are the world's youngest mountain range. They are prone to erosion and landslides. Seismic activity and rainstorms lash the region.

Real Estate Business

There is drastic increase of pilgrims who visit the holy places in the state annually. In the past decade, the number of tourists has risen by 155% (Uttarakhand Tourism Department).

A survey done by Indian Council for Research on International Economic Relation in 2006 states that there is an average of 102.5 hotels per million tourists. Shortage of dwelling units has led to mushrooming of illegal

Table-1: Increase in Pilgrims number

Yamunotri	Gangotri	Kedarnath	Badrinath
240%	250%	378%	136%
Increase of Tourists from 2001 to 2012	Increase of Tourists from 2001 to 2012	Increase of Tourists from 2001 to 2012	Increase of Tourists from 2001 to 2012
209, 753	252 783	323 867	489 924
Pilgrims in 2013 till June 20	Pilgrims in 2013 till June 20	Pilgrims in 2013 till June 20	Pilgrims in 2013 till June 20

structures, some right on the riverbanks, while there is the state government's notification of 2000 prohibits construction within 200 meters from the riverbanks. This limitation in 2002 was reduced to 100 meters of the river banks. Uttarakhand, High Court Bench comprising Chief Justice Barin Ghose and Justice Alok Singh ordered the State Govt. for demolition of all structures along the banks of Ganga, Song, Bhagirathi, Alaknanda and the Mandakini. But the state govt. did not act and floods have brought down hundreds of small hotels in the riverbanks.

Conversion of forest, land for Mining projects:

Secondly, statistics of the forest department shows that between 2000 and 2010, as many as, 3903.24 hectares (ha) forest land was diverted for mining projects. Besides, this illegal mining has hollowed down for boulders, pebbles, sand and gravel. In the meantime, state govt. changed the mining policy which facilitates auctioning of sites identified by the department of mining and geology. It helped the state govt. in generating profit of Rs. 300-350 crore. In 2012, additional 1608 ha forest land converted for mining.

Un-scientific mining helped rivers increase their width and change course. As per mining guidelines, not more than 0.9 meter should be dug, but private mining companies go as far down as a meters. During monsoon, when the river swells, it tilts towards the dug up area, not only changing its course but also putting roads and houses along the riverbank at high risk. In the last decade, the state government has diverted 15,072 ha Forest land for roads, irrigation, power transmission and hydel projects.

Table-2: Number of Hydel projects, Area of Riverbed Mining & conversion of Forest land

1	2	3	4	5	6	7
Almora	**Bageshwar**	**Champawat**	**Chamoli**	**Dehradun**	**Haridwar**	**Nainital**
Hydel Projects 1 Riverbed Mining 59.6 ha Forest Diverted 598.4 ha	Hydel Projects 13 Riverbed Mining 13.87 ha Forest Diverted 478.3 ha	Hydel Projects 2 Riverbed Mining 182.8 ha Forest Diverted 308.5 ha	Hydel Projects 51 Riverbed Mining 115.8 ha Forest Diverted 1,766.7 ha	Hydel Projects 11 Riverbed Mining 63.51 ha Forest Diverted 1,203.1 ha	Hydel Projects 2 Riverbed Mining - Forest Diverted 5,176 ha	Hydel Projects 4 Riverbed Mining 123.83 ha Forest Diverted 1,104.7 ha

8	9	10	11	12	13	
Pauri	**Pithoragarh**	**Rudraprayag**	**Tehri Garhwal**	**Uttarkashi**	**Udham Singh Nagar**	**Total**
Hydel Projects 13 Riverbed Mining 67.91 ha Forest Diverted 610.7 ha	Hydel Projects 62 Riverbed Mining 34.08 ha Forest Diverted 1,281.6 ha	Hydel Projects 19 Riverbed Mining 51.38 ha Forest Diverted 299 ha	Hydel Projects 23 Riverbed Mining 29.56 ha Forest Diverted 1,522 ha	Hydel Projects 42 Riverbed Mining 141.84 ha Forest Diverted 577.5 ha	Hydel Projects 1 Riverbed Mining 724.69 ha Forest Diverted 145.1 ha	**Hydel Projects 244 Riverbed Mining 1,608.9 ha Forest Diverted 15,072 ha**

Destruction by Dams

The Website of Uttarakhand Jal Vidyut Nigam Limited (UJVNL) shows that 45 Hydropower projects with a total capacity of 3164 MW are operational in Uttarakhand. The state plans to build 199 big and small projects. In the Alaknanda – Bhagirathi basin alone, 69 hydropower projects are built or proposed, states a report prepared in April 2013 by the environmental ministry's Inter-ministerial Group (IMG) formed to consider environmental flows and hydropower projects on the Ganga and its tributaries. Dam construction involves blasting, excavation, debris dumping, movement of heavy machinery, diversion of forests and rivers. This has a huge cumulative impact on Himalayan Ecology. The need is to make sustainable use of resources with minimal disturbance to ecology. In short, we may say "The way projects are being executed is disastrous."

Dams no doubt, are essential for meeting energy requirement but it is equally essential that the governments should also assess how much hydropower it actually needs. The 69 projects, when implemented would affect 81% of the Bhagirathi and 65% of the Alaknanda. Most of these are small projects which would reroute rivers, water through tunnels cut through the mountain, leaving long stretches of the rivers dry. Worse, the project proponents do not leave enough space (3-5 Km from each other) between two projects for the river to regenerate.

The problem aggravates because small projects do not require Environment Impact Assessment (EIA), so, cumulative impact of so many dams on a river can be horrendous. In view of a central water commission report that reservoirs on some rivers like Ramganga are overflowing with water 440% above the normal mark, scientists say there is a need to review dam building concepts such as Design Flood. This ensure that the dam can hold enough water to avert even a disastrous flood mark. But the Indian standard for fixing criteria for design flood for safety of dams does not directly incorporate the concept of hazard.

In Uttarakhand the situation aggravated this time because of indiscriminate dumping of debris and muck along the riverbanks. "This increases the erosive capacity of the river, which increases the river's water level." This causes extensive destruction in the downstream. Un-planned construction of hydropower projects also affects, rivers ecological flow, or the minimum water a river requires for its ecosystem and human needs.

Inter Ministerial Group (IMG) has recommended that between November and April, at least 30% of water of the rivers flow should be maintained. Between May and October, the monsoon season, ecological flow should be 20 per cent.

CONCLUSION

Disaster prevention is concerned with policies and programs to prevent the recurrence of natural disasters and covers the long term aspect of such disasters. The small price to pay for any method of prevention and protection pays off in the long. But after the catastrophic floods that caused widespread devastation in Uttarakhand on 16th June 2013 the state government seems to have learnt no lesson. The government seems to have forgotten to develop its own disaster management policy in the wake of repeated disaster striking the hill state since 2010 in form of floods, landslides and cloudbursts have created havoc in the state causing enormous loss to human lives and state's economy.

It appears that time has come to develop a common Himalayan policy involving not only India but other Himalayan boarder countries like China and Pakistan also. Himalayas, one of the youngest mountain range are very prone to erosion and landslides. Mindless development on this ecologically fragile mountain is one of the biggest reasons the flood have been so devastating in Uttarakhand, this time as "Nature has spoken, and more loudly this time."Secondly, it appears that "the river has come down to cleanse Uttarakhand of its greed."

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